

What is ICT?

Learning Objectives

- Define What is ICT? and explain its purpose
- Identify key components and features of What is ICT?
- Apply What is ICT? skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about What is ICT?? How is it used in real life?' Write student responses on the board. Guide discussion to connect What is ICT? to the chapter theme of Internet and Computing Basics.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of What is ICT?. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of What is ICT?. Define important terms and provide examples. Connect What is ICT? to the broader theme of Internet and Computing Basics. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to What is ICT?. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of What is ICT? through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Memory units use powers of 10 (KB, MB, GB) - place value

Science: Electricity powers computer components - circuits

Language: Labeling diagrams builds technical vocabulary

Digital Citizen Reminder: Handle computer equipment carefully - it is expensive and shared by all students.

Resources Needed:

- Computer diagram or real computer
- labeled chart
- exit ticket paper

Differentiation

Approaching: Match pictures of computer parts to their names using a word bank

On Level: Label the computer diagram from memory and explain each part

Advanced: Create a buying guide for a school computer lab - list parts and prices

SEND: Touch and identify real computer parts with teacher guidance

Formative Assessment

Method: Exit ticket and self-assessment

CT Skill Assessed: Decomposition - breaking down a computer into input, processing, memory, and output

Dormitory Task (Paper-and-Pencil)

Draw a labeled diagram of a computer from memory. Mark each part and write one sentence about what it does.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

History of Computers

Learning Objectives

- Define History of Computers and explain its purpose
- Identify key components and features of History of Computers
- Apply History of Computers skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about History of Computers? How is it used in real life?' Write student responses on the board. Guide discussion to connect History of Computers to the chapter theme of Internet and Computing Basics.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of History of Computers. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of History of Computers. Define important terms and provide examples. Connect History of Computers to the broader theme of Internet and Computing Basics. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to History of Computers. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of History of Computers through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Timeline as a number line - understanding centuries and decades

Science: Evolution of technology mirrors biological evolution

Language: Writing historical narratives builds chronological writing skills

Digital Citizen Reminder: Appreciate how computing has evolved - modern computers fit in your pocket.

Resources Needed:

- Timeline chart
- picture cards of old computers
- worksheet

Differentiation

Approaching: Match pictures of old and new computers to their correct time period

On Level: Create a timeline of computer evolution with 5 milestones

Advanced: Research and compare processing speeds of 1950s vs modern computers

SEND: Sort computer pictures from oldest to newest with teacher guidance

Formative Assessment

Method: Timeline activity and oral presentation

CT Skill Assessed: Pattern Recognition - identifying trends in computer evolution

Dormitory Task (Paper-and-Pencil)

Draw a timeline showing at least 4 generations of computers. Label each with the year and one key feature.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Introduction to Internet

Learning Objectives

- Define Introduction to Internet and explain its purpose
- Identify key components and features of Introduction to Internet
- Apply Introduction to Internet skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Introduction to Internet? How is it used in real life?' Write student responses on the board. Guide discussion to connect Introduction to Internet to the chapter theme of Internet and Computing Basics.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Introduction to Internet. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Introduction to Internet. Define important terms and provide examples. Connect Introduction to Internet to the broader theme of Internet and Computing Basics. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Introduction to Internet. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Introduction to Internet through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Network topology relates to graph theory in mathematics

Science: How data travels through cables and waves - physics of signals

Language: Writing about how the internet works builds expository writing

Digital Citizen Reminder: Never share personal information online - keep your full name, address, and school private.

Resources Needed:

- Internet diagram
- network map
- safety rules poster

Differentiation

Approaching: Match internet terms to their definitions using a word bank

On Level: Describe the journey of an email from sender to receiver

Advanced: Compare different internet connection types and their speeds

SEND: Identify parts of a network diagram with teacher guidance

Formative Assessment

Method: Diagram labeling and matching quiz

CT Skill Assessed: Decomposition - understanding how data moves from one computer to another

Dormitory Task (Paper-and-Pencil)

Draw a simple diagram showing how two computers connect through the internet. Label each part.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Web Browsers

Learning Objectives

- Define Web Browsers and explain its purpose
- Identify key components and features of Web Browsers
- Apply Web Browsers skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Web Browsers? How is it used in real life?' Write student responses on the board. Guide discussion to connect Web Browsers to the chapter theme of Internet and Computing Basics.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Web Browsers. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Web Browsers. Define important terms and provide examples. Connect Web Browsers to the broader theme of Internet and Computing Basics. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Web Browsers. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Web Browsers through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Understanding URL structure relates to pattern recognition

Science: Browsers render code into visual pages - computer science

Language: Writing clear instructions for browser use - procedural text

Digital Citizen Reminder: Keep your browser updated - updates fix security problems and add new features.

Resources Needed:

- Browser screenshot
- toolbar labeled diagram
- practice worksheet

Differentiation

Approaching: Identify browser toolbar buttons from a labeled diagram

On Level: Navigate to three different websites using the address bar

Advanced: Compare two different browsers and create a feature comparison table

SEND: Use browser features with teacher guidance - back, forward, refresh

Formative Assessment

Method: Practical navigation task

CT Skill Assessed: Algorithm Design - following steps to complete browser tasks

Dormitory Task (Paper-and-Pencil)

Draw a web browser window and label the address bar, back button, forward button, and refresh button.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Search Engines

Learning Objectives

- Define Search Engines and explain its purpose
- Identify key components and features of Search Engines
- Apply Search Engines skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Search Engines? How is it used in real life?' Write student responses on the board. Guide discussion to connect Search Engines to the chapter theme of Internet and Computing Basics.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Search Engines. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Search Engines. Define important terms and provide examples. Connect Search Engines to the broader theme of Internet and Computing Basics. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Search Engines. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Search Engines through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Search results ranked by relevance - understanding algorithms

Science: How search engines crawl and index the web - information science

Language: Writing effective search queries builds precise language skills

Digital Citizen Reminder: Not all search results are reliable - check if information comes from a trusted source.

Resources Needed:

- Computer with internet
- search tips handout
- worksheet

Differentiation

Approaching: Type simple search queries and click on results

On Level: Use three search techniques (quotes, minus, site:) to find specific information

Advanced: Evaluate search results for reliability and compare different sources

SEND: Perform a search with teacher guidance using step-by-step instructions

Formative Assessment

Method: Search skills practical test

CT Skill Assessed: Evaluation - assessing the quality and reliability of search results

Dormitory Task (Paper-and-Pencil)

Write five different search queries you would use to find information about your school. Explain which one would give the best results.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

URLs and Websites

Learning Objectives

- Define URLs and Websites and explain its purpose
- Identify key components and features of URLs and Websites
- Apply URLs and Websites skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about URLs and Websites? How is it used in real life?' Write student responses on the board. Guide discussion to connect URLs and Websites to the chapter theme of Internet and Computing Basics.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of URLs and Websites. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of URLs and Websites. Define important terms and provide examples. Connect URLs and Websites to the broader theme of Internet and Computing Basics. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to URLs and Websites. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of URLs and Websites through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: URL structure follows specific rules - understanding protocols and syntax

Science: Websites are hosted on servers located worldwide - geography

Language: Writing clear web content - audience awareness and structure

Digital Citizen Reminder: Check URLs carefully before clicking - fake websites can steal your information.

Resources Needed:

- URL anatomy diagram
- website examples
- labeling worksheet

Differentiation

Approaching: Identify the parts of a URL from a labeled diagram

On Level: Classify websites as educational, entertainment, or commercial

Advanced: Compare URLs to identify secure (https) vs non-secure connections

SEND: Read and identify URL components with teacher guidance

Formative Assessment

Method: URL labeling quiz

CT Skill Assessed: Abstraction - understanding that URLs follow a structured format

Dormitory Task (Paper-and-Pencil)

Write the URL of your favorite website. Label the protocol, domain name, and page name.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Introduction to GPS

Learning Objectives

- Define Introduction to GPS and explain its purpose
- Identify key components and features of Introduction to GPS
- Apply Introduction to GPS skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Introduction to GPS? How is it used in real life?' Write student responses on the board. Guide discussion to connect Introduction to GPS to the chapter theme of Internet and Computing Basics.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Introduction to GPS. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Introduction to GPS. Define important terms and provide examples. Connect Introduction to GPS to the broader theme of Internet and Computing Basics. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Introduction to GPS. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Introduction to GPS through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: GPS uses latitude and longitude coordinates - geography and math

Science: Satellites orbit Earth using principles of physics and gravity

Language: Writing directions builds spatial language and sequential writing

Digital Citizen Reminder: GPS tracks your location - be aware of privacy settings on your devices.

Resources Needed:

- GPS device or phone with GPS
- map
- coordinate worksheet

Differentiation

Approaching: Read latitude and longitude from a simple map

On Level: Use GPS coordinates to locate three places on a map

Advanced: Explain how GPS satellites determine your exact position

SEND: Find your location on a map with teacher guidance

Formative Assessment

Method: Coordinate reading activity

CT Skill Assessed: Decomposition - breaking down how GPS determines location using satellites

Dormitory Task (Paper-and-Pencil)

Write the latitude and longitude of your school. Explain in your own words how GPS finds this location.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Google Earth

Learning Objectives

- Define Google Earth and explain its purpose
- Identify key components and features of Google Earth
- Apply Google Earth skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Google Earth? How is it used in real life?' Write student responses on the board. Guide discussion to connect Google Earth to the chapter theme of Internet and Computing Basics.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Google Earth. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Google Earth. Define important terms and provide examples. Connect Google Earth to the broader theme of Internet and Computing Basics. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Google Earth. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Google Earth through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Understanding scale and zoom levels - mathematical ratios

Science: Satellite imagery and remote sensing - earth science

Language: Writing travel guides requires descriptive and persuasive writing

Digital Citizen Reminder: Respect privacy - do not zoom into private properties or share others' locations.

Resources Needed:

- Computer with Google Earth
- location coordinates
- exploration worksheet

Differentiation

Approaching: Navigate to your school using the search bar

On Level: Measure the distance between two cities using the ruler tool

Advanced: Create a virtual tour of five landmarks with descriptions

SEND: Explore locations with teacher guidance using basic navigation

Formative Assessment

Method: Virtual tour presentation

CT Skill Assessed: Evaluation - assessing the accuracy and usefulness of satellite imagery

Dormitory Task (Paper-and-Pencil)

Use Google Earth to find your school. Write the name of three nearby landmarks and estimate their distance from your school.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Internet Safety

Learning Objectives

- Define Internet Safety and explain its purpose
- Identify key components and features of Internet Safety
- Apply Internet Safety skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Internet Safety? How is it used in real life?' Write student responses on the board. Guide discussion to connect Internet Safety to the chapter theme of Internet and Computing Basics.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Internet Safety. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Internet Safety. Define important terms and provide examples. Connect Internet Safety to the broader theme of Internet and Computing Basics. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Internet Safety. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Internet Safety through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Risk assessment in math - probability and consequences

Science: Understanding online threats relates to biology (viruses as metaphor)

Language: Writing safety rules builds persuasive and instructional writing

Digital Citizen Reminder: Never share your password with anyone except your parents or teacher. Use strong passwords with letters, numbers, and symbols.

Resources Needed:

- Safety rules poster
- scenario cards
- pledge template

Differentiation

Approaching: Match online safety rules to their correct category

On Level: Identify safe and unsafe behaviors in five online scenarios

Advanced: Create a personal internet safety plan with at least 8 rules

SEND: Discuss safety rules with teacher guidance using picture prompts

Formative Assessment

Method: Safety pledge and scenario quiz

CT Skill Assessed: Evaluation - assessing online situations for safety risks

Dormitory Task (Paper-and-Pencil)

Write ten internet safety rules. For each rule, explain what could happen if you do not follow it.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Email Basics

Learning Objectives

- Define Email Basics and explain its purpose
- Identify key components and features of Email Basics
- Apply Email Basics skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Email Basics? How is it used in real life?' Write student responses on the board. Guide discussion to connect Email Basics to the chapter theme of Internet and Computing Basics.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Email Basics. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Email Basics. Define important terms and provide examples. Connect Email Basics to the broader theme of Internet and Computing Basics. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Email Basics. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Email Basics through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Understanding email structure - formal letter writing in language

Science: Email uses addresses like mathematical notation

Language: Writing clear subject lines and messages builds concise writing

Digital Citizen Reminder: Do not open emails from strangers - they may contain viruses or scams.

Resources Needed:

- Email template
- sample emails
- practice worksheet

Differentiation

Approaching: Read a sample email and identify the sender, subject, and body

On Level: Compose a simple email with subject line and greeting

Advanced: Write a formal email with proper structure, tone, and formatting

SEND: Send an email with teacher guidance - filling in To, Subject, and Body

Formative Assessment

Method: Email composition task

CT Skill Assessed: Algorithm Design - following the correct steps to compose and send an email

Dormitory Task (Paper-and-Pencil)

Write an email to your teacher asking for permission to use the computer lab. Include To, Subject, and body.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Downloading Files

Learning Objectives

- Define Downloading Files and explain its purpose
- Identify key components and features of Downloading Files
- Apply Downloading Files skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Downloading Files? How is it used in real life?' Write student responses on the board. Guide discussion to connect Downloading Files to the chapter theme of Internet and Computing Basics.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Downloading Files. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Downloading Files. Define important terms and provide examples. Connect Downloading Files to the broader theme of Internet and Computing Basics. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Downloading Files. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Downloading Files through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: File sizes in bytes, KB, MB - unit conversion and estimation

Science: Understanding how files travel through networks - physics

Language: Following download instructions builds procedural writing skills

Digital Citizen Reminder: Only download files from trusted websites. Scan all downloads for viruses before opening.

Resources Needed:

- Computer with internet
- download folder
- practice files

Differentiation

Approaching: Download a file and save it to the correct folder

On Level: Download three files and organize them by type

Advanced: Manage download settings and file storage efficiently

SEND: Download and save a file with teacher guidance step by step

Formative Assessment

Method: Practical download and organize task

CT Skill Assessed: Decomposition - breaking down the download process into steps

Dormitory Task (Paper-and-Pencil)

Write the steps to download a file from the internet. List at least 5 steps in order.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Chapter 1 Review

Learning Objectives

- Define Chapter 1 Review and explain its purpose
- Identify key components and features of Chapter 1 Review
- Apply Chapter 1 Review skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Chapter 1 Review? How is it used in real life?' Write student responses on the board. Guide discussion to connect Chapter 1 Review to the chapter theme of Internet and Computing Basics.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Chapter 1 Review. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Chapter 1 Review. Define important terms and provide examples. Connect Chapter 1 Review to the broader theme of Internet and Computing Basics. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Chapter 1 Review. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Chapter 1 Review through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Review exercises reinforce mathematical problem-solving

Science: Chapter review connects all internet and computing concepts

Language: Writing summaries builds synthesis and review skills

Digital Citizen Reminder: Apply internet safety rules you learned in every online activity.

Resources Needed:

- Review worksheet
- answer key
- chapter summary

Differentiation

Approaching: Complete fill-in-the-blank review questions

On Level: Answer short answer questions about chapter concepts

Advanced: Create a concept map connecting all chapter topics

SEND: Review key terms with teacher guidance using flashcards

Formative Assessment

Method: Chapter review quiz

CT Skill Assessed: Evaluation - self-assessing understanding of chapter concepts

Dormitory Task (Paper-and-Pencil)

Write a summary of everything you learned in this chapter. Include at least 8 key points.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Introduction to Graphics

Learning Objectives

- Define Introduction to Graphics and explain its purpose
- Identify key components and features of Introduction to Graphics
- Apply Introduction to Graphics skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Introduction to Graphics? How is it used in real life?' Write student responses on the board. Guide discussion to connect Introduction to Graphics to the chapter theme of Graphics.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Introduction to Graphics. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Introduction to Graphics. Define important terms and provide examples. Connect Introduction to Graphics to the broader theme of Graphics. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Introduction to Graphics. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Introduction to Graphics through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Pixel art uses grid coordinates - connecting to coordinate geometry

Science: Color mixing in digital art relates to light science

Language: Describing visual elements builds art vocabulary

Digital Citizen Reminder: Respect copyright - do not copy others artwork without permission.

Resources Needed:

- Computer with Paint
- color wheel
- sample graphics

Differentiation

Approaching: Identify basic shapes and colors in digital images

On Level: Create a simple drawing using at least 5 different tools

Advanced: Design a complex graphic combining multiple techniques

SEND: Use basic drawing tools with teacher guidance

Formative Assessment

Method: Tool identification and basic drawing task

CT Skill Assessed: Decomposition - breaking complex images into simple shapes

Dormitory Task (Paper-and-Pencil)

Draw a picture of your home using only basic shapes - circles, squares, triangles. Label each shape.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Paint Software Basics

Learning Objectives

- Define Paint Software Basics and explain its purpose
- Identify key components and features of Paint Software Basics
- Apply Paint Software Basics skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Paint Software Basics? How is it used in real life?' Write student responses on the board. Guide discussion to connect Paint Software Basics to the chapter theme of Graphics.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Paint Software Basics. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Paint Software Basics. Define important terms and provide examples. Connect Paint Software Basics to the broader theme of Graphics. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Paint Software Basics. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Paint Software Basics through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Understanding tool functions relates to following mathematical procedures

Science: Digital brushes simulate real art materials - materials science

Language: Writing tool guides builds instructional writing

Digital Citizen Reminder: Save your artwork frequently to avoid losing your work.

Resources Needed:

- Computer with Paint
- toolbar diagram
- practice template

Differentiation

Approaching: Use the pencil tool to draw a simple house

On Level: Create a picture using at least 4 different tools

Advanced: Design a landscape using multiple tools and techniques

SEND: Draw with different tools and identify their effects with teacher guidance

Formative Assessment

Method: Tool usage demonstration

CT Skill Assessed: Algorithm Design - following steps to create artwork

Dormitory Task (Paper-and-Pencil)

Write the names of five Paint tools and what each one does.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Drawing Tools

Learning Objectives

- Define Drawing Tools and explain its purpose
- Identify key components and features of Drawing Tools
- Apply Drawing Tools skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Drawing Tools? How is it used in real life?' Write student responses on the board. Guide discussion to connect Drawing Tools to the chapter theme of Graphics.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Drawing Tools. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Drawing Tools. Define important terms and provide examples. Connect Drawing Tools to the broader theme of Graphics. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Drawing Tools. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Drawing Tools through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Geometric shapes in art - identifying mathematical shapes in drawings

Science: Understanding brush strokes relates to fluid dynamics

Language: Describing drawing techniques builds technical vocabulary

Digital Citizen Reminder: Use drawing tools responsibly - do not draw on screens with markers or sharp objects.

Resources Needed:

- Computer with Paint
- tool practice sheets
- example artwork

Differentiation

Approaching: Use the line and rectangle tools to draw a building

On Level: Create a picture using at least 6 different drawing tools

Advanced: Combine advanced tools to create a detailed scene

SEND: Practice drawing with each tool and describe the results

Formative Assessment

Method: Drawing portfolio and tool checklist

CT Skill Assessed: Pattern Recognition - using repeated elements in designs

Dormitory Task (Paper-and-Pencil)

Draw a picture using only straight lines and rectangles. Count and write how many of each you used.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Color Selection

Learning Objectives

- Define Color Selection and explain its purpose
- Identify key components and features of Color Selection
- Apply Color Selection skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Color Selection? How is it used in real life?' Write student responses on the board. Guide discussion to connect Color Selection to the chapter theme of Graphics.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Color Selection. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Color Selection. Define important terms and provide examples. Connect Color Selection to the broader theme of Graphics. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Color Selection. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Color Selection through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Color theory - primary and secondary colors, mixing ratios

Science: How screens produce color using RGB light - physics

Language: Describing colors builds descriptive and sensory vocabulary

Digital Citizen Reminder: Choose appropriate colors for your artwork - some color combinations are hard to read.

Resources Needed:

- Color wheel
- color palette samples
- practice artwork

Differentiation

Approaching: Fill shapes with correct colors from a guide

On Level: Create a color palette and use it in a drawing

Advanced: Apply color theory to create mood and emphasis in artwork

SEND: Select and apply colors with teacher guidance

Formative Assessment

Method: Color matching activity

CT Skill Assessed: Abstraction - understanding that digital colors combine red, green, and blue

Dormitory Task (Paper-and-Pencil)

Draw a color wheel showing primary and secondary colors. Label each color.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Symmetric Designs

Learning Objectives

- Define Symmetric Designs and explain its purpose
- Identify key components and features of Symmetric Designs
- Apply Symmetric Designs skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Symmetric Designs? How is it used in real life?' Write student responses on the board. Guide discussion to connect Symmetric Designs to the chapter theme of Graphics.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Symmetric Designs. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Symmetric Designs. Define important terms and provide examples. Connect Symmetric Designs to the broader theme of Graphics. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Symmetric Designs. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Symmetric Designs through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Symmetry in math - reflection, rotation, and translation

Science: Symmetry in nature - biology and natural forms

Language: Writing about symmetry requires precise geometric language

Digital Citizen Reminder: Symmetry tools create mirror images - think before you use them.

Resources Needed:

- Computer with Paint
- symmetry examples
- practice grid

Differentiation

Approaching: Create a simple mirror image using the select and copy tools

On Level: Design a symmetric pattern with 4-fold rotational symmetry

Advanced: Create a complex mandala combining reflection and rotation

SEND: Make a symmetric design with teacher guidance step by step

Formative Assessment

Method: Symmetry design assessment

CT Skill Assessed: Pattern Recognition - identifying and creating symmetrical patterns

Dormitory Task (Paper-and-Pencil)

Fold a paper in half and draw half a butterfly. Open it to see the full symmetric design.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Image Editing

Learning Objectives

- Define Image Editing and explain its purpose
- Identify key components and features of Image Editing
- Apply Image Editing skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Image Editing? How is it used in real life?' Write student responses on the board. Guide discussion to connect Image Editing to the chapter theme of Graphics.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Image Editing. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Image Editing. Define important terms and provide examples. Connect Image Editing to the broader theme of Graphics. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Image Editing. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Image Editing through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Crop dimensions relate to area and perimeter calculations

Science: Image compression reduces file size - understanding ratios

Language: Writing editing instructions builds precise procedural text

Digital Citizen Reminder: Only edit images you have permission to use - respect copyright.

Resources Needed:

- Computer with image editor
- sample images
- editing checklist

Differentiation

Approaching: Crop an image to remove unwanted parts

On Level: Resize, crop, and adjust brightness of an image

Advanced: Apply multiple edits including crop, resize, rotate, and color adjust

SEND: Edit an image with teacher guidance using basic tools

Formative Assessment

Method: Image editing task

CT Skill Assessed: Algorithm Design - following a sequence of editing steps

Dormitory Task (Paper-and-Pencil)

Write a list of 5 image editing operations and explain what each one does.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Layers Concept

Learning Objectives

- Define Layers Concept and explain its purpose
- Identify key components and features of Layers Concept
- Apply Layers Concept skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Layers Concept? How is it used in real life?' Write student responses on the board. Guide discussion to connect Layers Concept to the chapter theme of Graphics.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Layers Concept. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Layers Concept. Define important terms and provide examples. Connect Layers Concept to the broader theme of Graphics. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Layers Concept. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Layers Concept through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Layers stack like fractions - understanding ordering and precedence

Science: How transparent layers work - optics and light

Language: Explaining layers requires clear organizational writing

Digital Citizen Reminder: Understanding layers helps organize work - a skill useful beyond computing.

Resources Needed:

- Computer with layer-capable editor
- layer examples
- practice files

Differentiation

Approaching: Create two layers and draw different elements on each

On Level: Use layers to create a composite image with foreground and background

Advanced: Manage complex multi-layer compositions with blending modes

SEND: Create and manage layers with teacher guidance

Formative Assessment

Method: Layer management demonstration

CT Skill Assessed: Abstraction - understanding that layers separate elements for independent editing

Dormitory Task (Paper-and-Pencil)

Draw three separate elements (sun, house, tree) on separate pieces of paper. Stack them to create one picture.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Filters and Effects

Learning Objectives

- Define Filters and Effects and explain its purpose
- Identify key components and features of Filters and Effects
- Apply Filters and Effects skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Filters and Effects? How is it used in real life?' Write student responses on the board. Guide discussion to connect Filters and Effects to the chapter theme of Graphics.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Filters and Effects. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Filters and Effects. Define important terms and provide examples. Connect Filters and Effects to the broader theme of Graphics. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Filters and Effects. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Filters and Effects through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Filters transform images mathematically - understanding transformations

Science: Digital filters mimic photographic techniques - optics

Language: Describing visual effects builds descriptive vocabulary

Digital Citizen Reminder: Effects should enhance your work, not hide poor quality - aim for quality over effects.

Resources Needed:

- Computer with image editor
- filter gallery
- before-after examples

Differentiation

Approaching: Apply one filter and compare before and after

On Level: Apply three different effects and describe what each does

Advanced: Combine multiple effects to create an artistic style

SEND: Apply and compare filters with teacher guidance

Formative Assessment

Method: Effect application and comparison

CT Skill Assessed: Evaluation - assessing which effects improve an image

Dormitory Task (Paper-and-Pencil)

Write the names of four image effects and describe what each one does to a picture.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Text in Graphics

Learning Objectives

- Define Text in Graphics and explain its purpose
- Identify key components and features of Text in Graphics
- Apply Text in Graphics skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Text in Graphics? How is it used in real life?' Write student responses on the board. Guide discussion to connect Text in Graphics to the chapter theme of Graphics.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Text in Graphics. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Text in Graphics. Define important terms and provide examples. Connect Text in Graphics to the broader theme of Graphics. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Text in Graphics. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Text in Graphics through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Text in Graphics - geometric transformations

Science: Scientific applications of Text in Graphics - digital imaging

Language: Writing about Text in Graphics builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply vector graphics and image editing skills responsibly and ethically.

Resources Needed:

- Computer with image editor
- sample images
- tool reference

Differentiation

Approaching: Complete guided practice activities with vector graphics and image editing support

On Level: Demonstrate understanding of Text in Graphics concepts through practical exercises

Advanced: Apply Text in Graphics concepts to solve complex problems independently

SEND: Practice Text in Graphics skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Text in Graphics activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Text in Graphics into manageable steps

Dormitory Task (Paper-and-Pencil)

Draw a simple design using only basic shapes. Label the tools you would use to create it digitally.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

File Formats

Learning Objectives

- Define File Formats and explain its purpose
- Identify key components and features of File Formats
- Apply File Formats skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about File Formats? How is it used in real life?' Write student responses on the board. Guide discussion to connect File Formats to the chapter theme of Graphics.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of File Formats. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of File Formats. Define important terms and provide examples. Connect File Formats to the broader theme of Graphics. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to File Formats. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of File Formats through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in File Formats - geometric transformations

Science: Scientific applications of File Formats - digital imaging

Language: Writing about File Formats builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply vector graphics and image editing skills responsibly and ethically.

Resources Needed:

- Advanced Graphics reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with vector graphics and image editing support

On Level: Demonstrate understanding of File Formats concepts through practical exercises

Advanced: Apply File Formats concepts to solve complex problems independently

SEND: Practice File Formats skills with step-by-step teacher guidance

Formative Assessment

Method: Practical File Formats activity and oral questioning

CT Skill Assessed: Decomposition - breaking down File Formats into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about File Formats. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Project - School Logo

Learning Objectives

- Define Project - School Logo and explain its purpose
- Identify key components and features of Project - School Logo
- Apply Project - School Logo skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Project - School Logo? How is it used in real life?' Write student responses on the board. Guide discussion to connect Project - School Logo to the chapter theme of Graphics.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Project - School Logo. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Project - School Logo. Define important terms and provide examples. Connect Project - School Logo to the broader theme of Graphics. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Project - School Logo. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Project - School Logo through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Project - School Logo - geometric transformations

Science: Scientific applications of Project - School Logo - digital imaging

Language: Writing about Project - School Logo builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply vector graphics and image editing skills responsibly and ethically.

Resources Needed:

- Advanced Graphics reference materials
- project planning template
- self-assessment rubric

Differentiation

Approaching: Complete guided practice activities with vector graphics and image editing support

On Level: Demonstrate understanding of Project - School Logo concepts through practical exercises

Advanced: Apply Project - School Logo concepts to solve complex problems independently

SEND: Practice Project - School Logo skills with step-by-step teacher guidance

Formative Assessment

Method: Project rubric and self-assessment

CT Skill Assessed: Evaluation - assessing the quality and completeness of the Project - School Logo project

Dormitory Task (Paper-and-Pencil)

Plan a Project - School Logo project. Write the steps you would follow and what materials you need.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Chapter 2 Review

Learning Objectives

- Define Chapter 2 Review and explain its purpose
- Identify key components and features of Chapter 2 Review
- Apply Chapter 2 Review skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Chapter 2 Review? How is it used in real life?' Write student responses on the board. Guide discussion to connect Chapter 2 Review to the chapter theme of Graphics.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Chapter 2 Review. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Chapter 2 Review. Define important terms and provide examples. Connect Chapter 2 Review to the broader theme of Graphics. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Chapter 2 Review. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Chapter 2 Review through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Review exercises consolidate programming concepts

Science: Chapter summaries connect related topics

Language: Writing reviews builds synthesis skills

Digital Citizen Reminder: Apply Chapter 2 Review skills responsibly and ethically.

Resources Needed:

- Computer with Python
- code reference card
- practice exercises

Differentiation

Approaching: Complete guided practice with Chapter 2 Review support

On Level: Demonstrate Chapter 2 Review through practical exercises

Advanced: Apply Chapter 2 Review to solve complex problems independently

SEND: Practice Chapter 2 Review with step-by-step teacher guidance

Formative Assessment

Method: Practical Chapter 2 Review activity

CT Skill Assessed: Decomposition - breaking down Chapter 2 Review into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a simple Python program on paper that demonstrates Chapter 2 Review. Include comments.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Introduction to Word Processing

Learning Objectives

- Define Introduction to Word Processing and explain its purpose
- Identify key components and features of Introduction to Word Processing
- Apply Introduction to Word Processing skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Introduction to Word Processing? How is it used in real life?' Write student responses on the board. Guide discussion to connect Introduction to Word Processing to the chapter theme of Word Processing.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Introduction to Word Processing. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Introduction to Word Processing. Define important terms and provide examples. Connect Introduction to Word Processing to the broader theme of Word Processing. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Introduction to Word Processing. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Introduction to Word Processing through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Types of Communication - network topology

Science: Scientific applications of Types of Communication - data transmission

Language: Writing about Types of Communication builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply network configuration skills responsibly and ethically.

Resources Needed:

- Networking reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with network configuration support

On Level: Demonstrate understanding of Types of Communication concepts through practical exercises

Advanced: Apply Types of Communication concepts to solve complex problems independently

SEND: Practice Types of Communication skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Types of Communication activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Types of Communication into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Types of Communication. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Typing Skills

Learning Objectives

- Define Typing Skills and explain its purpose
- Identify key components and features of Typing Skills
- Apply Typing Skills skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Typing Skills? How is it used in real life?' Write student responses on the board. Guide discussion to connect Typing Skills to the chapter theme of Word Processing.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Typing Skills. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Typing Skills. Define important terms and provide examples. Connect Typing Skills to the broader theme of Word Processing. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Typing Skills. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Typing Skills through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Audio Communication - network topology

Science: Scientific applications of Audio Communication - data transmission

Language: Writing about Audio Communication builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply network configuration skills responsibly and ethically.

Resources Needed:

- Computer with media software
- sample media files
- editing guide

Differentiation

Approaching: Complete guided practice activities with network configuration support

On Level: Demonstrate understanding of Audio Communication concepts through practical exercises

Advanced: Apply Audio Communication concepts to solve complex problems independently

SEND: Practice Audio Communication skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Audio Communication activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Audio Communication into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a plan for a short audio or video project. List the steps and tools you would use.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Text Formatting

Learning Objectives

- Define Text Formatting and explain its purpose
- Identify key components and features of Text Formatting
- Apply Text Formatting skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Text Formatting? How is it used in real life?' Write student responses on the board. Guide discussion to connect Text Formatting to the chapter theme of Word Processing.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Text Formatting. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Text Formatting. Define important terms and provide examples. Connect Text Formatting to the broader theme of Word Processing. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Text Formatting. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Text Formatting through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Video Communication - network topology

Science: Scientific applications of Video Communication - data transmission

Language: Writing about Video Communication builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply network configuration skills responsibly and ethically.

Resources Needed:

- Computer with media software
- sample media files
- editing guide

Differentiation

Approaching: Complete guided practice activities with network configuration support

On Level: Demonstrate understanding of Video Communication concepts through practical exercises

Advanced: Apply Video Communication concepts to solve complex problems independently

SEND: Practice Video Communication skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Video Communication activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Video Communication into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a plan for a short audio or video project. List the steps and tools you would use.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Paragraph Formatting

Learning Objectives

- Define Paragraph Formatting and explain its purpose
- Identify key components and features of Paragraph Formatting
- Apply Paragraph Formatting skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Paragraph Formatting? How is it used in real life?' Write student responses on the board. Guide discussion to connect Paragraph Formatting to the chapter theme of Word Processing.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Paragraph Formatting. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Paragraph Formatting. Define important terms and provide examples. Connect Paragraph Formatting to the broader theme of Word Processing. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Paragraph Formatting. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Paragraph Formatting through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Introduction to Audacity - network topology

Science: Scientific applications of Introduction to Audacity - data transmission

Language: Writing about Introduction to Audacity builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply network configuration skills responsibly and ethically.

Resources Needed:

- Computer with media software
- sample media files
- editing guide

Differentiation

Approaching: Complete guided practice activities with network configuration support

On Level: Demonstrate understanding of Introduction to Audacity concepts through practical exercises

Advanced: Apply Introduction to Audacity concepts to solve complex problems independently

SEND: Practice Introduction to Audacity skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Introduction to Audacity activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Introduction to Audacity into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Introduction to Audacity. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Tables in Documents

Learning Objectives

- Define Tables in Documents and explain its purpose
- Identify key components and features of Tables in Documents
- Apply Tables in Documents skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Tables in Documents? How is it used in real life?' Write student responses on the board. Guide discussion to connect Tables in Documents to the chapter theme of Word Processing.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Tables in Documents. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Tables in Documents. Define important terms and provide examples. Connect Tables in Documents to the broader theme of Word Processing. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Tables in Documents. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Tables in Documents through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Recording Audio - network topology

Science: Scientific applications of Recording Audio - data transmission

Language: Writing about Recording Audio builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply network configuration skills responsibly and ethically.

Resources Needed:

- Computer with media software
- sample media files
- editing guide

Differentiation

Approaching: Complete guided practice activities with network configuration support

On Level: Demonstrate understanding of Recording Audio concepts through practical exercises

Advanced: Apply Recording Audio concepts to solve complex problems independently

SEND: Practice Recording Audio skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Recording Audio activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Recording Audio into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a plan for a short audio or video project. List the steps and tools you would use.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Mail Merge

Learning Objectives

- Define Mail Merge and explain its purpose
- Identify key components and features of Mail Merge
- Apply Mail Merge skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Mail Merge? How is it used in real life?' Write student responses on the board. Guide discussion to connect Mail Merge to the chapter theme of Word Processing.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Mail Merge. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Mail Merge. Define important terms and provide examples. Connect Mail Merge to the broader theme of Word Processing. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Mail Merge. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Mail Merge through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Editing Audio - network topology

Science: Scientific applications of Editing Audio - data transmission

Language: Writing about Editing Audio builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply network configuration skills responsibly and ethically.

Resources Needed:

- Computer with media software
- sample media files
- editing guide

Differentiation

Approaching: Complete guided practice activities with network configuration support

On Level: Demonstrate understanding of Editing Audio concepts through practical exercises

Advanced: Apply Editing Audio concepts to solve complex problems independently

SEND: Practice Editing Audio skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Editing Audio activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Editing Audio into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a plan for a short audio or video project. List the steps and tools you would use.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Reports and Projects

Learning Objectives

- Define Reports and Projects and explain its purpose
- Identify key components and features of Reports and Projects
- Apply Reports and Projects skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Reports and Projects? How is it used in real life?' Write student responses on the board. Guide discussion to connect Reports and Projects to the chapter theme of Word Processing.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Reports and Projects. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Reports and Projects. Define important terms and provide examples. Connect Reports and Projects to the broader theme of Word Processing. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Reports and Projects. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Reports and Projects through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Effects and Filters - network topology

Science: Scientific applications of Effects and Filters - data transmission

Language: Writing about Effects and Filters builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply network configuration skills responsibly and ethically.

Resources Needed:

- Computer with image editor
- sample images
- tool reference

Differentiation

Approaching: Complete guided practice activities with network configuration support

On Level: Demonstrate understanding of Effects and Filters concepts through practical exercises

Advanced: Apply Effects and Filters concepts to solve complex problems independently

SEND: Practice Effects and Filters skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Effects and Filters activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Effects and Filters into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Effects and Filters. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Letters and Applications

Learning Objectives

- Define Letters and Applications and explain its purpose
- Identify key components and features of Letters and Applications
- Apply Letters and Applications skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Letters and Applications? How is it used in real life?' Write student responses on the board. Guide discussion to connect Letters and Applications to the chapter theme of Word Processing.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Letters and Applications. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Letters and Applications. Define important terms and provide examples. Connect Letters and Applications to the broader theme of Word Processing. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Letters and Applications. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Letters and Applications through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Mixing Audio - network topology

Science: Scientific applications of Mixing Audio - data transmission

Language: Writing about Mixing Audio builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply network configuration skills responsibly and ethically.

Resources Needed:

- Computer with media software
- sample media files
- editing guide

Differentiation

Approaching: Complete guided practice activities with network configuration support

On Level: Demonstrate understanding of Mixing Audio concepts through practical exercises

Advanced: Apply Mixing Audio concepts to solve complex problems independently

SEND: Practice Mixing Audio skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Mixing Audio activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Mixing Audio into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a plan for a short audio or video project. List the steps and tools you would use.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Advanced Formatting

Learning Objectives

- Define Advanced Formatting and explain its purpose
- Identify key components and features of Advanced Formatting
- Apply Advanced Formatting skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Advanced Formatting? How is it used in real life?' Write student responses on the board. Guide discussion to connect Advanced Formatting to the chapter theme of Word Processing.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Advanced Formatting. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Advanced Formatting. Define important terms and provide examples. Connect Advanced Formatting to the broader theme of Word Processing. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Advanced Formatting. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Advanced Formatting through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Exporting Audio - network topology

Science: Scientific applications of Exporting Audio - data transmission

Language: Writing about Exporting Audio builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply network configuration skills responsibly and ethically.

Resources Needed:

- Computer with media software
- sample media files
- editing guide

Differentiation

Approaching: Complete guided practice activities with network configuration support

On Level: Demonstrate understanding of Exporting Audio concepts through practical exercises

Advanced: Apply Exporting Audio concepts to solve complex problems independently

SEND: Practice Exporting Audio skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Exporting Audio activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Exporting Audio into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a plan for a short audio or video project. List the steps and tools you would use.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Spreadsheets Basics

Learning Objectives

- Define Spreadsheets Basics and explain its purpose
- Identify key components and features of Spreadsheets Basics
- Apply Spreadsheets Basics skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Spreadsheets Basics? How is it used in real life?' Write student responses on the board. Guide discussion to connect Spreadsheets Basics to the chapter theme of Word Processing.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Spreadsheets Basics. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Spreadsheets Basics. Define important terms and provide examples. Connect Spreadsheets Basics to the broader theme of Word Processing. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Spreadsheets Basics. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Spreadsheets Basics through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Video Editing Basics - network topology

Science: Scientific applications of Video Editing Basics - data transmission

Language: Writing about Video Editing Basics builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply network configuration skills responsibly and ethically.

Resources Needed:

- Computer with media software
- sample media files
- editing guide

Differentiation

Approaching: Complete guided practice activities with network configuration support

On Level: Demonstrate understanding of Video Editing Basics concepts through practical exercises

Advanced: Apply Video Editing Basics concepts to solve complex problems independently

SEND: Practice Video Editing Basics skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Video Editing Basics activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Video Editing Basics into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a plan for a short audio or video project. List the steps and tools you would use.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Presentations Basics

Learning Objectives

- Define Presentations Basics and explain its purpose
- Identify key components and features of Presentations Basics
- Apply Presentations Basics skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Presentations Basics? How is it used in real life?' Write student responses on the board. Guide discussion to connect Presentations Basics to the chapter theme of Word Processing.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Presentations Basics. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Presentations Basics. Define important terms and provide examples. Connect Presentations Basics to the broader theme of Word Processing. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Presentations Basics. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Presentations Basics through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Project - Audio Story - network topology

Science: Scientific applications of Project - Audio Story - data transmission

Language: Writing about Project - Audio Story builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply network configuration skills responsibly and ethically.

Resources Needed:

- Networking reference materials
- project planning template
- self-assessment rubric

Differentiation

Approaching: Complete guided practice activities with network configuration support

On Level: Demonstrate understanding of Project - Audio Story concepts through practical exercises

Advanced: Apply Project - Audio Story concepts to solve complex problems independently

SEND: Practice Project - Audio Story skills with step-by-step teacher guidance

Formative Assessment

Method: Project rubric and self-assessment

CT Skill Assessed: Evaluation - assessing the quality and completeness of the Project - Audio Story project

Dormitory Task (Paper-and-Pencil)

Plan a Project - Audio Story project. Write the steps you would follow and what materials you need.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Chapter 3 Review

Learning Objectives

- Define Chapter 3 Review and explain its purpose
- Identify key components and features of Chapter 3 Review
- Apply Chapter 3 Review skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Chapter 3 Review? How is it used in real life?' Write student responses on the board. Guide discussion to connect Chapter 3 Review to the chapter theme of Word Processing.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Chapter 3 Review. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Chapter 3 Review. Define important terms and provide examples. Connect Chapter 3 Review to the broader theme of Word Processing. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Chapter 3 Review. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Chapter 3 Review through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Review exercises consolidate database concepts

Science: Chapter summaries connect SQL topics

Language: Writing reviews builds synthesis skills

Digital Citizen Reminder: Apply Chapter 3 Review skills responsibly and ethically.

Resources Needed:

- Computer with Python
- code reference card
- practice exercises

Differentiation

Approaching: Complete guided practice with Chapter 3 Review support

On Level: Demonstrate Chapter 3 Review through practical exercises

Advanced: Apply Chapter 3 Review to solve complex problems independently

SEND: Practice Chapter 3 Review with step-by-step teacher guidance

Formative Assessment

Method: Practical Chapter 3 Review activity

CT Skill Assessed: Decomposition - breaking down Chapter 3 Review into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a simple Python program on paper that demonstrates Chapter 3 Review. Include comments.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Introduction to Spreadsheets

Learning Objectives

- Define Introduction to Spreadsheets and explain its purpose
- Identify key components and features of Introduction to Spreadsheets
- Apply Introduction to Spreadsheets skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Introduction to Spreadsheets? How is it used in real life?' Write student responses on the board. Guide discussion to connect Introduction to Spreadsheets to the chapter theme of Spreadsheets.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Introduction to Spreadsheets. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Introduction to Spreadsheets. Define important terms and provide examples. Connect Introduction to Spreadsheets to the broader theme of Spreadsheets. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Introduction to Spreadsheets. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Introduction to Spreadsheets through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Introduction to Data - mathematical concepts

Science: Scientific applications of Introduction to Data - scientific applications

Language: Writing about Introduction to Data builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills responsibly and ethically.

Resources Needed:

- Chapter 4 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Introduction to Data concepts through practical exercises

Advanced: Apply Introduction to Data concepts to solve complex problems independently

SEND: Practice Introduction to Data skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Introduction to Data activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Introduction to Data into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Introduction to Data. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Formulas and Functions

Learning Objectives

- Define Formulas and Functions and explain its purpose
- Identify key components and features of Formulas and Functions
- Apply Formulas and Functions skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Formulas and Functions? How is it used in real life?' Write student responses on the board. Guide discussion to connect Formulas and Functions to the chapter theme of Spreadsheets.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Formulas and Functions. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Formulas and Functions. Define important terms and provide examples. Connect Formulas and Functions to the broader theme of Spreadsheets. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Formulas and Functions. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Formulas and Functions through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Number Systems - mathematical concepts

Science: Scientific applications of Number Systems - scientific applications

Language: Writing about Number Systems builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills responsibly and ethically.

Resources Needed:

- Chapter 4 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Number Systems concepts through practical exercises

Advanced: Apply Number Systems concepts to solve complex problems independently

SEND: Practice Number Systems skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Number Systems activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Number Systems into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Number Systems. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Charts and Graphs

Learning Objectives

- Define Charts and Graphs and explain its purpose
- Identify key components and features of Charts and Graphs
- Apply Charts and Graphs skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Charts and Graphs? How is it used in real life?' Write student responses on the board. Guide discussion to connect Charts and Graphs to the chapter theme of Spreadsheets.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Charts and Graphs. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Charts and Graphs. Define important terms and provide examples. Connect Charts and Graphs to the broader theme of Spreadsheets. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Charts and Graphs. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Charts and Graphs through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Binary Conversion - mathematical concepts

Science: Scientific applications of Binary Conversion - scientific applications

Language: Writing about Binary Conversion builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:

- Chapter 4 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Binary Conversion concepts through practical exercises

Advanced: Apply Binary Conversion concepts to solve complex problems independently

SEND: Practice Binary Conversion skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Binary Conversion activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Binary Conversion into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Binary Conversion. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Data Sorting and Filtering

Learning Objectives

- Define Data Sorting and Filtering and explain its purpose
- Identify key components and features of Data Sorting and Filtering
- Apply Data Sorting and Filtering skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Data Sorting and Filtering? How is it used in real life?' Write student responses on the board. Guide discussion to connect Data Sorting and Filtering to the chapter theme of Spreadsheets.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Data Sorting and Filtering. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Data Sorting and Filtering. Define important terms and provide examples. Connect Data Sorting and Filtering to the broader theme of Spreadsheets. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Data Sorting and Filtering. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Data Sorting and Filtering through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Text Representation - mathematical concepts

Science: Scientific applications of Text Representation - scientific applications

Language: Writing about Text Representation builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills responsibly and ethically.

Resources Needed:

- Computer with presentation software
- design guide
- sample slides

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Text Representation concepts through practical exercises

Advanced: Apply Text Representation concepts to solve complex problems independently

SEND: Practice Text Representation skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Text Representation activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Text Representation into manageable steps

Dormitory Task (Paper-and-Pencil)

Plan a 4-slide presentation about Text Representation. Write the title and 3 bullet points for each slide.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Conditional Formatting

Learning Objectives

- Define Conditional Formatting and explain its purpose
- Identify key components and features of Conditional Formatting
- Apply Conditional Formatting skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Conditional Formatting? How is it used in real life?' Write student responses on the board. Guide discussion to connect Conditional Formatting to the chapter theme of Spreadsheets.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Conditional Formatting. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Conditional Formatting. Define important terms and provide examples. Connect Conditional Formatting to the broader theme of Spreadsheets. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Conditional Formatting. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Conditional Formatting through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Image Representation - mathematical concepts

Science: Scientific applications of Image Representation - scientific applications

Language: Writing about Image Representation builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills responsibly and ethically.

Resources Needed:

- Computer with image editor
- sample images
- tool reference

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Image Representation concepts through practical exercises

Advanced: Apply Image Representation concepts to solve complex problems independently

SEND: Practice Image Representation skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Image Representation activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Image Representation into manageable steps

Dormitory Task (Paper-and-Pencil)

Draw a simple design using only basic shapes. Label the tools you would use to create it digitally.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Budget and Accounts

Learning Objectives

- Define Budget and Accounts and explain its purpose
- Identify key components and features of Budget and Accounts
- Apply Budget and Accounts skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Budget and Accounts? How is it used in real life?' Write student responses on the board. Guide discussion to connect Budget and Accounts to the chapter theme of Spreadsheets.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Budget and Accounts. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Budget and Accounts. Define important terms and provide examples. Connect Budget and Accounts to the broader theme of Spreadsheets. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Budget and Accounts. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Budget and Accounts through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Audio Representation - mathematical concepts

Science: Scientific applications of Audio Representation - scientific applications

Language: Writing about Audio Representation builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills responsibly and ethically.

Resources Needed:

- Computer with media software
- sample media files
- editing guide

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Audio Representation concepts through practical exercises

Advanced: Apply Audio Representation concepts to solve complex problems independently

SEND: Practice Audio Representation skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Audio Representation activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Audio Representation into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a plan for a short audio or video project. List the steps and tools you would use.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Student Marks Tracker

Learning Objectives

- Define Student Marks Tracker and explain its purpose
- Identify key components and features of Student Marks Tracker
- Apply Student Marks Tracker skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Student Marks Tracker? How is it used in real life?' Write student responses on the board. Guide discussion to connect Student Marks Tracker to the chapter theme of Spreadsheets.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Student Marks Tracker. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Student Marks Tracker. Define important terms and provide examples. Connect Student Marks Tracker to the broader theme of Spreadsheets. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Student Marks Tracker. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Student Marks Tracker through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Data processing transforms raw data into useful information

Science: Processing pipelines connect multiple operations

Language: Writing data processing code builds ETL skills

Digital Citizen Reminder: Apply Data Processing skills responsibly and ethically.

Resources Needed:

- Computer with Python
- code reference card
- practice exercises

Differentiation

Approaching: Complete guided practice with Data Processing support

On Level: Demonstrate Data Processing through practical exercises

Advanced: Apply Data Processing to solve complex problems independently

SEND: Practice Data Processing with step-by-step teacher guidance

Formative Assessment

Method: Practical Data Processing activity

CT Skill Assessed: Decomposition - breaking down Data Processing into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a simple Python program on paper that demonstrates Data Processing. Include comments.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Data Analysis

Learning Objectives

- Define Data Analysis and explain its purpose
- Identify key components and features of Data Analysis
- Apply Data Analysis skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Data Analysis? How is it used in real life?' Write student responses on the board. Guide discussion to connect Data Analysis to the chapter theme of Spreadsheets.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Data Analysis. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Data Analysis. Define important terms and provide examples. Connect Data Analysis to the broader theme of Spreadsheets. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Data Analysis. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Data Analysis through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Data Storage - mathematical concepts

Science: Scientific applications of Data Storage - scientific applications

Language: Writing about Data Storage builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:

- Chapter 4 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Data Storage concepts through practical exercises

Advanced: Apply Data Storage concepts to solve complex problems independently

SEND: Practice Data Storage skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Data Storage activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Data Storage into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Data Storage. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Spreadsheet Project

Learning Objectives

- Define Spreadsheet Project and explain its purpose
- Identify key components and features of Spreadsheet Project
- Apply Spreadsheet Project skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Spreadsheet Project? How is it used in real life?' Write student responses on the board. Guide discussion to connect Spreadsheet Project to the chapter theme of Spreadsheets.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Spreadsheet Project. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Spreadsheet Project. Define important terms and provide examples. Connect Spreadsheet Project to the broader theme of Spreadsheets. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Spreadsheet Project. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Spreadsheet Project through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Data Compression - mathematical concepts

Science: Scientific applications of Data Compression - scientific applications

Language: Writing about Data Compression builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills responsibly and ethically.

Resources Needed:

- Chapter 4 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Data Compression concepts through practical exercises

Advanced: Apply Data Compression concepts to solve complex problems independently

SEND: Practice Data Compression skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Data Compression activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Data Compression into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Data Compression. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Charts Advanced

Learning Objectives

- Define Charts Advanced and explain its purpose
- Identify key components and features of Charts Advanced
- Apply Charts Advanced skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Charts Advanced? How is it used in real life?' Write student responses on the board. Guide discussion to connect Charts Advanced to the chapter theme of Spreadsheets.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Charts Advanced. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Charts Advanced. Define important terms and provide examples. Connect Charts Advanced to the broader theme of Spreadsheets. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Charts Advanced. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Charts Advanced through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Data Encryption - mathematical concepts

Science: Scientific applications of Data Encryption - scientific applications

Language: Writing about Data Encryption builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:

- Chapter 4 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Data Encryption concepts through practical exercises

Advanced: Apply Data Encryption concepts to solve complex problems independently

SEND: Practice Data Encryption skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Data Encryption activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Data Encryption into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Data Encryption. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Data Validation

Learning Objectives

- Define Data Validation and explain its purpose
- Identify key components and features of Data Validation
- Apply Data Validation skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Data Validation? How is it used in real life?' Write student responses on the board. Guide discussion to connect Data Validation to the chapter theme of Spreadsheets.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Data Validation. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Data Validation. Define important terms and provide examples. Connect Data Validation to the broader theme of Spreadsheets. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Data Validation. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Data Validation through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Project - Data Analysis - mathematical concepts

Science: Scientific applications of Project - Data Analysis - scientific applications

Language: Writing about Project - Data Analysis builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills responsibly and ethically.

Resources Needed:

- Chapter 4 reference materials
- project planning template
- self-assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Project - Data Analysis concepts through practical exercises

Advanced: Apply Project - Data Analysis concepts to solve complex problems independently

SEND: Practice Project - Data Analysis skills with step-by-step teacher guidance

Formative Assessment

Method: Project rubric and self-assessment

CT Skill Assessed: Evaluation - assessing the quality and completeness of the Project - Data Analysis project

Dormitory Task (Paper-and-Pencil)

Plan a Project - Data Analysis project. Write the steps you would follow and what materials you need.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Macros Introduction

Learning Objectives

- Define Macros Introduction and explain its purpose
- Identify key components and features of Macros Introduction
- Apply Macros Introduction skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Macros Introduction? How is it used in real life?' Write student responses on the board. Guide discussion to connect Macros Introduction to the chapter theme of Spreadsheets.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Macros Introduction. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Macros Introduction. Define important terms and provide examples. Connect Macros Introduction to the broader theme of Spreadsheets. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Macros Introduction. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Macros Introduction through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Review exercises consolidate web development concepts

Science: Chapter summaries connect HTML and CSS topics

Language: Writing reviews builds synthesis skills

Digital Citizen Reminder: Apply Chapter 4 Review skills responsibly and ethically.

Resources Needed:

- Computer with Python
- code reference card
- practice exercises

Differentiation

Approaching: Complete guided practice with Chapter 4 Review support

On Level: Demonstrate Chapter 4 Review through practical exercises

Advanced: Apply Chapter 4 Review to solve complex problems independently

SEND: Practice Chapter 4 Review with step-by-step teacher guidance

Formative Assessment

Method: Practical Chapter 4 Review activity

CT Skill Assessed: Decomposition - breaking down Chapter 4 Review into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a simple Python program on paper that demonstrates Chapter 4 Review. Include comments.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Introduction to Presentations

Learning Objectives

- Define Introduction to Presentations and explain its purpose
- Identify key components and features of Introduction to Presentations
- Apply Introduction to Presentations skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Introduction to Presentations? How is it used in real life?' Write student responses on the board. Guide discussion to connect Introduction to Presentations to the chapter theme of Presentations.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Introduction to Presentations. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Introduction to Presentations. Define important terms and provide examples. Connect Introduction to Presentations to the broader theme of Presentations. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Introduction to Presentations. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Introduction to Presentations through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Block-based programming follows logical sequences - algorithm concepts

Science: Scratch coordinates use x,y plane - connecting to coordinate geometry

Language: Writing Scratch programs builds computational thinking

Digital Citizen Reminder: Share your Scratch projects responsibly and give credit to others.

Resources Needed:

- Computer with Scratch
- project worksheet
- design template

Differentiation

Approaching: Move a sprite to three positions using three move blocks

On Level: Create an animation with a sprite moving and changing costume

Advanced: Create a story with multiple sprites, backdrops, and dialogue

SEND: Click and drag blocks with teacher guidance to move a sprite

Formative Assessment

Method: Project checklist and demonstration

CT Skill Assessed: Decomposition - breaking an animation into individual sprite actions

Dormitory Task (Paper-and-Pencil)

Draw a 4-panel comic strip showing a character moving from left to right. Label what happens in each panel.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Slide Design Principles

Learning Objectives

- Define Slide Design Principles and explain its purpose
- Identify key components and features of Slide Design Principles
- Apply Slide Design Principles skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Slide Design Principles? How is it used in real life?' Write student responses on the board. Guide discussion to connect Slide Design Principles to the chapter theme of Presentations.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Slide Design Principles. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Slide Design Principles. Define important terms and provide examples. Connect Slide Design Principles to the broader theme of Presentations. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Slide Design Principles. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Slide Design Principles through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Motion Blocks - mathematical concepts

Science: Scientific applications of Motion Blocks - scientific applications

Language: Writing about Motion Blocks builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills responsibly and ethically.

Resources Needed:

- Computer with Scratch
- block reference card
- project worksheet

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Motion Blocks concepts through practical exercises

Advanced: Apply Motion Blocks concepts to solve complex problems independently

SEND: Practice Motion Blocks skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Motion Blocks activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Motion Blocks into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Motion Blocks. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Animations and Transitions

Learning Objectives

- Define Animations and Transitions and explain its purpose
- Identify key components and features of Animations and Transitions
- Apply Animations and Transitions skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Animations and Transitions? How is it used in real life?' Write student responses on the board. Guide discussion to connect Animations and Transitions to the chapter theme of Presentations.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Animations and Transitions. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Animations and Transitions. Define important terms and provide examples. Connect Animations and Transitions to the broader theme of Presentations. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Animations and Transitions. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Animations and Transitions through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Looks Blocks - mathematical concepts

Science: Scientific applications of Looks Blocks - scientific applications

Language: Writing about Looks Blocks builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills responsibly and ethically.

Resources Needed:

- Computer with Scratch
- block reference card
- project worksheet

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Looks Blocks concepts through practical exercises

Advanced: Apply Looks Blocks concepts to solve complex problems independently

SEND: Practice Looks Blocks skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Looks Blocks activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Looks Blocks into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Looks Blocks. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Speaker Notes and Delivery

Learning Objectives

- Define Speaker Notes and Delivery and explain its purpose
- Identify key components and features of Speaker Notes and Delivery
- Apply Speaker Notes and Delivery skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Speaker Notes and Delivery? How is it used in real life?' Write student responses on the board. Guide discussion to connect Speaker Notes and Delivery to the chapter theme of Presentations.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Speaker Notes and Delivery. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Speaker Notes and Delivery. Define important terms and provide examples. Connect Speaker Notes and Delivery to the broader theme of Presentations. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Speaker Notes and Delivery. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Speaker Notes and Delivery through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Sound Blocks - mathematical concepts

Science: Scientific applications of Sound Blocks - scientific applications

Language: Writing about Sound Blocks builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills responsibly and ethically.

Resources Needed:

- Computer with Scratch
- block reference card
- project worksheet

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Sound Blocks concepts through practical exercises

Advanced: Apply Sound Blocks concepts to solve complex problems independently

SEND: Practice Sound Blocks skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Sound Blocks activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Sound Blocks into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Sound Blocks. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Collaborative Presentations

Learning Objectives

- Define Collaborative Presentations and explain its purpose
- Identify key components and features of Collaborative Presentations
- Apply Collaborative Presentations skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Collaborative Presentations? How is it used in real life?' Write student responses on the board. Guide discussion to connect Collaborative Presentations to the chapter theme of Presentations.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Collaborative Presentations. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Collaborative Presentations. Define important terms and provide examples. Connect Collaborative Presentations to the broader theme of Presentations. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Collaborative Presentations. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Collaborative Presentations through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Event handling uses triggers - cause and effect in science

Science: Control structures (loops, conditionals) follow logical rules

Language: Writing event-driven programs builds sequential thinking

Digital Citizen Reminder: Test your programs frequently - finding errors early saves time.

Resources Needed:

- Computer with Scratch
- events reference
- practice exercises

Differentiation

Approaching: Use the green flag event to start a program

On Level: Combine events with loops for continuous animation

Advanced: Create a complex program using multiple events and conditional logic

SEND: Use the green flag and repeat block with teacher guidance

Formative Assessment

Method: Event handling demonstration

CT Skill Assessed: Algorithm Design - creating programs with events and control structures

Dormitory Task (Paper-and-Pencil)

Write the steps for a program that makes a sprite move when you click the green flag.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Presentation Review

Learning Objectives

- Define Presentation Review and explain its purpose
- Identify key components and features of Presentation Review
- Apply Presentation Review skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Presentation Review? How is it used in real life?' Write student responses on the board. Guide discussion to connect Presentation Review to the chapter theme of Presentations.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Presentation Review. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Presentation Review. Define important terms and provide examples. Connect Presentation Review to the broader theme of Presentations. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Presentation Review. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Presentation Review through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Variables - mathematical concepts

Science: Scientific applications of Variables - scientific applications

Language: Writing about Variables builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills responsibly and ethically.

Resources Needed:

- Chapter 5 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Variables concepts through practical exercises

Advanced: Apply Variables concepts to solve complex problems independently

SEND: Practice Variables skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Variables activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Variables into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Variables. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Presentations Project

Learning Objectives

- Define Presentations Project and explain its purpose
- Identify key components and features of Presentations Project
- Apply Presentations Project skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Presentations Project? How is it used in real life?' Write student responses on the board. Guide discussion to connect Presentations Project to the chapter theme of Presentations.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Presentations Project. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Presentations Project. Define important terms and provide examples. Connect Presentations Project to the broader theme of Presentations. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Presentations Project. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Presentations Project through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: If-else statements use Boolean logic - true/false conditions

Science: Comparison operators (>, <, =) are mathematical

Language: Writing conditions builds logical expression skills

Digital Citizen Reminder: Test all possible conditions - programs should handle unexpected inputs.

Resources Needed:

- Computer with Scratch
- conditionals reference
- practice exercises

Differentiation

Approaching: Use if-else to change sprite color on key press

On Level: Create a simple game with if-else for scoring

Advanced: Build a complex game with nested conditions and multiple decision paths

SEND: Use an if statement with teacher guidance

Formative Assessment

Method: Conditional logic demonstration

CT Skill Assessed: Algorithm Design - creating programs with decision-making

Dormitory Task (Paper-and-Pencil)

Write three if-else conditions for a game. Example: If score > 10, display You Win, else display Keep Trying.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Loops

Learning Objectives

- Define Loops and explain its purpose
- Identify key components and features of Loops
- Apply Loops skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Loops? How is it used in real life?' Write student responses on the board. Guide discussion to connect Loops to the chapter theme of Presentations.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Loops. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Loops. Define important terms and provide examples. Connect Loops to the broader theme of Presentations. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Loops. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Loops through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Loops - mathematical concepts

Science: Scientific applications of Loops - scientific applications

Language: Writing about Loops builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills responsibly and ethically.

Resources Needed:

- Computer with Python
- code reference card
- practice exercises

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Loops concepts through practical exercises

Advanced: Apply Loops concepts to solve complex problems independently

SEND: Practice Loops skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Loops activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Loops into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Loops. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Sensing

Learning Objectives

- Define Sensing and explain its purpose
- Identify key components and features of Sensing
- Apply Sensing skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Sensing? How is it used in real life?' Write student responses on the board. Guide discussion to connect Sensing to the chapter theme of Presentations.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Sensing. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Sensing. Define important terms and provide examples. Connect Sensing to the broader theme of Presentations. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Sensing. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Sensing through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Sensing - mathematical concepts

Science: Scientific applications of Sensing - scientific applications

Language: Writing about Sensing builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills responsibly and ethically.

Resources Needed:

- Chapter 5 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Sensing concepts through practical exercises

Advanced: Apply Sensing concepts to solve complex problems independently

SEND: Practice Sensing skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Sensing activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Sensing into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Sensing. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Operators

Learning Objectives

- Define Operators and explain its purpose
- Identify key components and features of Operators
- Apply Operators skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Operators? How is it used in real life?' Write student responses on the board. Guide discussion to connect Operators to the chapter theme of Presentations.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Operators. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Operators. Define important terms and provide examples. Connect Operators to the broader theme of Presentations. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Operators. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Operators through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Operators - mathematical concepts

Science: Scientific applications of Operators - scientific applications

Language: Writing about Operators builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills responsibly and ethically.

Resources Needed:

- Chapter 5 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Operators concepts through practical exercises

Advanced: Apply Operators concepts to solve complex problems independently

SEND: Practice Operators skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Operators activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Operators into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Operators. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Project - Animation

Learning Objectives

- Define Project - Animation and explain its purpose
- Identify key components and features of Project - Animation
- Apply Project - Animation skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Project - Animation? How is it used in real life?' Write student responses on the board. Guide discussion to connect Project - Animation to the chapter theme of Presentations.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Project - Animation. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Project - Animation. Define important terms and provide examples. Connect Project - Animation to the broader theme of Presentations. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Project - Animation. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Project - Animation through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Project - Animation - mathematical concepts

Science: Scientific applications of Project - Animation - scientific applications

Language: Writing about Project - Animation builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills responsibly and ethically.

Resources Needed:

- Chapter 5 reference materials
- project planning template
- self-assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Project - Animation concepts through practical exercises

Advanced: Apply Project - Animation concepts to solve complex problems independently

SEND: Practice Project - Animation skills with step-by-step teacher guidance

Formative Assessment

Method: Project rubric and self-assessment

CT Skill Assessed: Evaluation - assessing the quality and completeness of the Project - Animation project

Dormitory Task (Paper-and-Pencil)

Plan a Project - Animation project. Write the steps you would follow and what materials you need.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Chapter 5 Review

Learning Objectives

- Define Chapter 5 Review and explain its purpose
- Identify key components and features of Chapter 5 Review
- Apply Chapter 5 Review skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Chapter 5 Review? How is it used in real life?' Write student responses on the board. Guide discussion to connect Chapter 5 Review to the chapter theme of Presentations.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Chapter 5 Review. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Chapter 5 Review. Define important terms and provide examples. Connect Chapter 5 Review to the broader theme of Presentations. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Chapter 5 Review. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Chapter 5 Review through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Review exercises consolidate algorithm concepts

Science: Chapter summaries connect all algorithm topics

Language: Writing reviews builds synthesis skills

Digital Citizen Reminder: Apply Chapter 5 Review skills responsibly and ethically.

Resources Needed:

- Computer with Python
- code reference card
- practice exercises

Differentiation

Approaching: Complete guided practice with Chapter 5 Review support

On Level: Demonstrate Chapter 5 Review through practical exercises

Advanced: Apply Chapter 5 Review to solve complex problems independently

SEND: Practice Chapter 5 Review with step-by-step teacher guidance

Formative Assessment

Method: Practical Chapter 5 Review activity

CT Skill Assessed: Decomposition - breaking down Chapter 5 Review into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a simple Python program on paper that demonstrates Chapter 5 Review. Include comments.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Types of Communication

Learning Objectives

- Define Types of Communication and explain its purpose
- Identify key components and features of Types of Communication
- Apply Types of Communication skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Types of Communication? How is it used in real life?' Write student responses on the board. Guide discussion to connect Types of Communication to the chapter theme of Communication.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Types of Communication. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Types of Communication. Define important terms and provide examples. Connect Types of Communication to the broader theme of Communication. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Types of Communication. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Types of Communication through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in What is Software? - mathematical concepts

Science: Scientific applications of What is Software? - scientific applications

Language: Writing about What is Software? builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills responsibly and ethically.

Resources Needed:

- Chapter 6 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of What is Software? concepts through practical exercises

Advanced: Apply What is Software? concepts to solve complex problems independently

SEND: Practice What is Software? skills with step-by-step teacher guidance

Formative Assessment

Method: Practical What is Software? activity and oral questioning

CT Skill Assessed: Decomposition - breaking down What is Software? into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about What is Software?. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Audio Communication

Learning Objectives

- Define Audio Communication and explain its purpose
- Identify key components and features of Audio Communication
- Apply Audio Communication skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Audio Communication? How is it used in real life?' Write student responses on the board. Guide discussion to connect Audio Communication to the chapter theme of Communication.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Audio Communication. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Audio Communication. Define important terms and provide examples. Connect Audio Communication to the broader theme of Communication. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Audio Communication. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Audio Communication through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in System Software - mathematical concepts

Science: Scientific applications of System Software - scientific applications

Language: Writing about System Software builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills responsibly and ethically.

Resources Needed:

- Chapter 6 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of System Software concepts through practical exercises

Advanced: Apply System Software concepts to solve complex problems independently

SEND: Practice System Software skills with step-by-step teacher guidance

Formative Assessment

Method: Practical System Software activity and oral questioning

CT Skill Assessed: Decomposition - breaking down System Software into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about System Software. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Video Communication

Learning Objectives

- Define Video Communication and explain its purpose
- Identify key components and features of Video Communication
- Apply Video Communication skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Video Communication? How is it used in real life?' Write student responses on the board. Guide discussion to connect Video Communication to the chapter theme of Communication.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Video Communication. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Video Communication. Define important terms and provide examples. Connect Video Communication to the broader theme of Communication. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Video Communication. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Video Communication through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Application Software - mathematical concepts

Science: Scientific applications of Application Software - scientific applications

Language: Writing about Application Software builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills responsibly and ethically.

Resources Needed:

- Chapter 6 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Application Software concepts through practical exercises

Advanced: Apply Application Software concepts to solve complex problems independently

SEND: Practice Application Software skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Application Software activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Application Software into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Application Software. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Introduction to Audacity

Learning Objectives

- Define Introduction to Audacity and explain its purpose
- Identify key components and features of Introduction to Audacity
- Apply Introduction to Audacity skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Introduction to Audacity? How is it used in real life?' Write student responses on the board. Guide discussion to connect Introduction to Audacity to the chapter theme of Communication.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Introduction to Audacity. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Introduction to Audacity. Define important terms and provide examples. Connect Introduction to Audacity to the broader theme of Communication. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Introduction to Audacity. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Introduction to Audacity through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Word Processing - mathematical concepts

Science: Scientific applications of Word Processing - scientific applications

Language: Writing about Word Processing builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills responsibly and ethically.

Resources Needed:

- Chapter 6 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Word Processing concepts through practical exercises

Advanced: Apply Word Processing concepts to solve complex problems independently

SEND: Practice Word Processing skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Word Processing activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Word Processing into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Word Processing. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Recording Audio

Learning Objectives

- Define Recording Audio and explain its purpose
- Identify key components and features of Recording Audio
- Apply Recording Audio skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Recording Audio? How is it used in real life?' Write student responses on the board. Guide discussion to connect Recording Audio to the chapter theme of Communication.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Recording Audio. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Recording Audio. Define important terms and provide examples. Connect Recording Audio to the broader theme of Communication. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Recording Audio. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Recording Audio through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Spreadsheet Basics - mathematical concepts

Science: Scientific applications of Spreadsheet Basics - scientific applications

Language: Writing about Spreadsheet Basics builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:

- Computer with spreadsheet
- practice data set
- formula reference

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Spreadsheet Basics concepts through practical exercises

Advanced: Apply Spreadsheet Basics concepts to solve complex problems independently

SEND: Practice Spreadsheet Basics skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Spreadsheet Basics activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Spreadsheet Basics into manageable steps

Dormitory Task (Paper-and-Pencil)

Draw a simple spreadsheet on graph paper with 5 columns and 6 rows. Fill in sample data.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Editing Audio

Learning Objectives

- Define Editing Audio and explain its purpose
- Identify key components and features of Editing Audio
- Apply Editing Audio skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Editing Audio? How is it used in real life?' Write student responses on the board. Guide discussion to connect Editing Audio to the chapter theme of Communication.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Editing Audio. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Editing Audio. Define important terms and provide examples. Connect Editing Audio to the broader theme of Communication. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Editing Audio. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Editing Audio through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Presentation Software - mathematical concepts

Science: Scientific applications of Presentation Software - scientific applications

Language: Writing about Presentation Software builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:

- Computer with presentation software
- design guide
- sample slides

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Presentation Software concepts through practical exercises

Advanced: Apply Presentation Software concepts to solve complex problems independently

SEND: Practice Presentation Software skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Presentation Software activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Presentation Software into manageable steps

Dormitory Task (Paper-and-Pencil)

Plan a 4-slide presentation about Presentation Software. Write the title and 3 bullet points for each slide.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Effects and Filters

Learning Objectives

- Define Effects and Filters and explain its purpose
- Identify key components and features of Effects and Filters
- Apply Effects and Filters skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Effects and Filters? How is it used in real life?' Write student responses on the board. Guide discussion to connect Effects and Filters to the chapter theme of Communication.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Effects and Filters. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Effects and Filters. Define important terms and provide examples. Connect Effects and Filters to the broader theme of Communication. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Effects and Filters. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Effects and Filters through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Database Introduction - mathematical concepts

Science: Scientific applications of Database Introduction - scientific applications

Language: Writing about Database Introduction builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills responsibly and ethically.

Resources Needed:

- Database software
- practice data set
- query reference card

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Database Introduction concepts through practical exercises

Advanced: Apply Database Introduction concepts to solve complex problems independently

SEND: Practice Database Introduction skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Database Introduction activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Database Introduction into manageable steps

Dormitory Task (Paper-and-Pencil)

Design a simple database table for storing student information. List at least 5 fields with their data types.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Mixing Audio

Learning Objectives

- Define Mixing Audio and explain its purpose
- Identify key components and features of Mixing Audio
- Apply Mixing Audio skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Mixing Audio? How is it used in real life?' Write student responses on the board. Guide discussion to connect Mixing Audio to the chapter theme of Communication.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Mixing Audio. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Mixing Audio. Define important terms and provide examples. Connect Mixing Audio to the broader theme of Communication. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Mixing Audio. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Mixing Audio through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Utility Software - mathematical concepts

Science: Scientific applications of Utility Software - scientific applications

Language: Writing about Utility Software builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills responsibly and ethically.

Resources Needed:

- Chapter 6 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Utility Software concepts through practical exercises

Advanced: Apply Utility Software concepts to solve complex problems independently

SEND: Practice Utility Software skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Utility Software activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Utility Software into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Utility Software. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Exporting Audio

Learning Objectives

- Define Exporting Audio and explain its purpose
- Identify key components and features of Exporting Audio
- Apply Exporting Audio skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Exporting Audio? How is it used in real life?' Write student responses on the board. Guide discussion to connect Exporting Audio to the chapter theme of Communication.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Exporting Audio. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Exporting Audio. Define important terms and provide examples. Connect Exporting Audio to the broader theme of Communication. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Exporting Audio. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Exporting Audio through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Software Installation - mathematical concepts

Science: Scientific applications of Software Installation - scientific applications

Language: Writing about Software Installation builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills responsibly and ethically.

Resources Needed:

- Chapter 6 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Software Installation concepts through practical exercises

Advanced: Apply Software Installation concepts to solve complex problems independently

SEND: Practice Software Installation skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Software Installation activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Software Installation into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Software Installation. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Video Editing Basics

Learning Objectives

- Define Video Editing Basics and explain its purpose
- Identify key components and features of Video Editing Basics
- Apply Video Editing Basics skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Video Editing Basics? How is it used in real life?' Write student responses on the board. Guide discussion to connect Video Editing Basics to the chapter theme of Communication.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Video Editing Basics. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Video Editing Basics. Define important terms and provide examples. Connect Video Editing Basics to the broader theme of Communication. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Video Editing Basics. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Video Editing Basics through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Software Licenses - mathematical concepts

Science: Scientific applications of Software Licenses - scientific applications

Language: Writing about Software Licenses builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills responsibly and ethically.

Resources Needed:

- Chapter 6 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Software Licenses concepts through practical exercises

Advanced: Apply Software Licenses concepts to solve complex problems independently

SEND: Practice Software Licenses skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Software Licenses activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Software Licenses into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Software Licenses. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Project - Audio Story

Learning Objectives

- Define Project - Audio Story and explain its purpose
- Identify key components and features of Project - Audio Story
- Apply Project - Audio Story skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Project - Audio Story? How is it used in real life?' Write student responses on the board. Guide discussion to connect Project - Audio Story to the chapter theme of Communication.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Project - Audio Story. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Project - Audio Story. Define important terms and provide examples. Connect Project - Audio Story to the broader theme of Communication. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Project - Audio Story. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Project - Audio Story through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Project - Newsletter - mathematical concepts

Science: Scientific applications of Project - Newsletter - scientific applications

Language: Writing about Project - Newsletter builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills responsibly and ethically.

Resources Needed:

- Chapter 6 reference materials
- project planning template
- self-assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Project - Newsletter concepts through practical exercises

Advanced: Apply Project - Newsletter concepts to solve complex problems independently

SEND: Practice Project - Newsletter skills with step-by-step teacher guidance

Formative Assessment

Method: Project rubric and self-assessment

CT Skill Assessed: Evaluation - assessing the quality and completeness of the Project - Newsletter project

Dormitory Task (Paper-and-Pencil)

Plan a Project - Newsletter project. Write the steps you would follow and what materials you need.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Chapter 6 Review

Learning Objectives

- Define Chapter 6 Review and explain its purpose
- Identify key components and features of Chapter 6 Review
- Apply Chapter 6 Review skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Chapter 6 Review? How is it used in real life?' Write student responses on the board. Guide discussion to connect Chapter 6 Review to the chapter theme of Communication.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Chapter 6 Review. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Chapter 6 Review. Define important terms and provide examples. Connect Chapter 6 Review to the broader theme of Communication. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Chapter 6 Review. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Chapter 6 Review through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Review exercises consolidate technology concepts

Science: Chapter summaries connect all emerging tech topics

Language: Writing reviews builds synthesis skills

Digital Citizen Reminder: Apply Chapter 6 Review skills responsibly and ethically.

Resources Needed:

- Computer with Python
- code reference card
- practice exercises

Differentiation

Approaching: Complete guided practice with Chapter 6 Review support

On Level: Demonstrate Chapter 6 Review through practical exercises

Advanced: Apply Chapter 6 Review to solve complex problems independently

SEND: Practice Chapter 6 Review with step-by-step teacher guidance

Formative Assessment

Method: Practical Chapter 6 Review activity

CT Skill Assessed: Decomposition - breaking down Chapter 6 Review into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a simple Python program on paper that demonstrates Chapter 6 Review. Include comments.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

What is ICT?

Learning Objectives

- Define ICT and identify its components
- Recognize ICT devices used in daily life
- Understand the importance of ICT in modern society

Regional Metaphor: Grain Sorting as Data Processing

Teacher Activity

Ask students: “How do you sort grains at home? What steps do you follow?”

Guide discussion to connect grain sorting to data processing:

- Unsorted grains = Raw data
- Sorting process = Data processing
- Sorted grains = Information

Student Activity

Share their experiences of helping at home with grain sorting.

Give examples of different types of grains they know.

Materials Needed:

- Real grains (rice, wheat) for demonstration
- Chart paper and markers
- ICT textbook

Understanding ICT

Engage (5-7 minutes)

Teacher Activity

Show pictures of ICT devices: computer, mobile phone, tablet, television.

Ask: “How are these devices similar? How are they different?”

Student Activity

Observe the pictures and discuss in pairs.

Share observations with the class.

Explore (10-12 minutes)

Teacher Activity

Explain ICT = Information and Communication Technology

Show the components of ICT:

- Information: Data, text, images, audio, video
- Communication: Sharing, transmitting, receiving
- Technology: Tools, devices, software

Student Activity

Take notes in their notebooks.

Identify ICT components in their daily life.

Explain (10-12 minutes)

Teacher Activity

Live demonstration: Show how a computer processes information.

Use the grain sorting metaphor:

1. Input: Raw grains (unsorted)
2. Processing: Sorting process
3. Output: Sorted grains
4. Storage: Container to store sorted grains

Student Activity

Follow the demonstration.

Draw a diagram showing input → processing → output → storage.

Elaborate (8-10 minutes)

Teacher Activity

Assign Parson's Problem: Arrange the following steps in correct order:

- "We get sorted grains"
- "We put grains in the machine"
- "Machine sorts the grains"
- "We store sorted grains in container"

Student Activity

Work in pairs to arrange the steps.

Present their answer to the class.

Evaluate (5-8 minutes)

Teacher Activity

Conduct exit ticket: Ask students to write:

1. Full form of ICT
2. Two examples of ICT devices
3. One way ICT is useful in daily life

Student Activity

Complete the exit ticket.

- Submit before leaving the class.

Cross-Curricular Connections

Mathematics: Numbers and place value connect to binary representation

Science: Sound waves carry telephone signals

Language: Writing clear instructions mirrors algorithm design

Digital Citizen Reminder: Never share your password with anyone except your parents or teacher.

Dormitory Task (Paper-and-Pencil)

Complete the paper-and-pencil activity related to today's topic. Practice the concept without needing a computer.

Resources Needed:

- Chalkboard
- worksheet
- examples of ICT devices (if available)

Differentiation

Approaching: Use word bank with pictures

On Level: Core activity as described

Advanced: Research an ICT device at home

SEND: Paired support, visual aids

Formative Assessment

Method: Exit ticket and self-assessment

CT Skill Assessed: Decomposition — breaking down tasks into smaller parts

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson: